

READING PASSAGE 3

The History of Astronomy and Navigation

A In recent years, the history of astronomy and navigation has attracted considerable attention from both the academic community and the general public. This growing interest reflects a broader recognition of the importance of this topic for understanding some of the most pressing challenges facing contemporary society.

B Historically, scholarly understanding of this topic was limited by both theoretical frameworks and available methodologies. Early researchers operated within relatively narrow conceptual boundaries, and their findings, while groundbreaking for their time, often failed to capture the full complexity of the phenomena they sought to explain. The development of new research tools and analytical techniques in the latter half of the twentieth century fundamentally transformed the landscape, enabling scholars to investigate questions that had previously been considered beyond the reach of empirical inquiry.

C Contemporary research has identified several critical factors that determine outcomes in this domain. The interplay between environmental conditions, technological capabilities, and social organisation creates a complex adaptive system in which small changes can produce disproportionately large effects. Understanding these dynamics requires analytical approaches that move beyond simple cause-and-effect reasoning to embrace the principles of complexity science and systems thinking.

D The empirical record is further strengthened by a series of meta-analyses that have synthesised findings from hundreds of individual studies. These comprehensive reviews have consistently identified moderate to large effect sizes, with the most robust findings demonstrating remarkable consistency across different research designs, measurement approaches, and analytical strategies. This convergence of evidence from multiple independent lines of inquiry considerably strengthens confidence in the validity of the core theoretical propositions.

E The practical application of research findings has proven more challenging than many initially anticipated. The translation of academic knowledge into actionable policy recommendations requires navigating a complex landscape of political constraints, economic interests, and social dynamics that academic models often simplify or overlook. Implementation failures in several high-profile initiatives have underscored the gap between theoretical understanding and practical capacity, prompting calls for more engaged and context-sensitive approaches to research.

F The implications of this body of research extend well beyond the academy. As societies grapple with increasingly complex challenges, the insights generated by scholarly inquiry in this domain have the potential to inform more effective and equitable policy responses. Realising this potential, however, will require sustained investment in both fundamental research and the translational infrastructure needed to convert knowledge into action.

G In conclusion, the study of the history of astronomy and navigation represents a vital and dynamic area of contemporary research. The progress that has been made in understanding the key mechanisms and dynamics provides a solid foundation for both further scholarly inquiry and practical application. The challenges that remain should be seen not as reasons for discouragement but as opportunities for continued intellectual growth and social contribution.

Question 1

Choose the correct letter, A, B, C or D.

1 What does the passage identify as a significant challenge to current understanding?

- A. too much data collection
- B. the generalisability of findings across contexts
- C. insufficient government funding
- D. overly complex statistical methods

Question 2

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

2 The author notes that researchers have called for more _____ approaches to investigation.

Question 3

Choose the correct letter, A, B, C or D.

3 Which of the following best describes the author's view on future developments?

- A. pessimism about the future
- B. cautious optimism with acknowledgment of challenges
- C. complete certainty about outcomes
- D. indifference to practical applications

Question 4

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

4 The landmark study analysed data spanning a period of _____ years.

Question 5

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

5 All critics agree that the research findings are universally applicable.

Question 6

Choose the correct letter, A, B, C or D.

6 The meta-analyses cited in the passage reveal that:

- A. weak effects across studies
- B. moderate to large effect sizes with consistency
- C. contradictory findings
- D. results only in specific populations

Question 7

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 7 Policy-makers have fully implemented all research recommendations.

Questions 8–9

Complete the sentences below. Choose NO MORE THAN THREE WORDS from the passage for each answer.

- 8 According to the text, cognitive processes shape the _____ that people make.
- 9 According to the passage, structural factors include economic incentives, institutional frameworks, and _____.

Question 10

Choose the correct letter, A, B, C or D.

- 10 The meta-analyses cited in the passage reveal that:
- A. economic incentives and institutional constraints
 - B. individual personality traits
 - C. random chance
 - D. government policies alone

Questions 11–12

Do the following statements agree with the information given in the passage?

Write TRUE if the statement agrees with the information

Write FALSE if the statement contradicts the information

Write NOT GIVEN if there is no information on this

- 11 Early researchers had access to the same technologies as modern scholars.
- 12 The landmark study discussed in the passage involved participants from multiple countries.

Question 13

Choose the correct letter, A, B, C or D.

- 13 The author suggests that early studies in this field were limited by:
- A. weak effects across studies
 - B. moderate to large effect sizes with consistency
 - C. contradictory findings
 - D. results only in specific populations